Functionals

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1 Introduction

In mathematics, a functional is a mapping whose domain is a set of functions and whose codomain is usually the real numbers ($\mathbb{R}$) or the complex numbers ($\mathbb{C}$). In other words, it is a function of a function: it assigns a number to each admissible function:

$$F: X \to \mathbb{R} \quad \text{or} \quad F: X \to \mathbb{C}$$

where X is a function space, such as a space of continuous or square-integrable functions. The domain must be chosen so that each expression is well defined. Examples of functionals:

1. Point-Evaluation Functional

$$F[f] = f(x_0)$$

Assigns to f its value at a fixed point x_0.

2. Integral Functional

$$F[f] = \int_a^b f\,(x)\,dx$$

Assigns to f its definite integral over $[a, b]$.

3. Norm Functional

$$F[f] = \left(\int_a^b |f(x)|^2\,dx\right)^{1/2}$$

Assigns to f its norm in $L^2[a, b]$. A function f belongs to the space L^2 if the integral of the square of its absolute value is a finite real number:

$$\int_X |f(x)|^2 dx \ < \infty.$$

where X represents the space of functions.

4. Simple Integral Functional

$$F[f] = \int_a^b f\,(x)\,dx$$

Returns the signed area under $f(x)$ over $[a, b]$. Example: calculating the work done by a variable force $F(x)$ along a displacement.

5. Time-Integrated Mechanical Energy

$$E[y] = \int_{t_1}^{t_2} \left(\frac{1}{2}m\dot{y}(t)^2 + \frac{1}{2}ky(t)^2\right) dt$$

Assigns to each trajectory $y(t)$ the time integral of its kinetic plus potential energy over $[t_1, t_2]$. Example: a mass attached to a spring. This quantity has dimensions of energy times time; it is not the instantaneous total mechanical energy or the action, whose integrand is kinetic minus potential energy.

6. Action Functional and the Stationary-Action Principle

$$S[y] = \int_{t_1}^{t_2} L\,(y(t), \dot{y}(t), t)\,dt$$

Here, L is the Lagrangian. Example: determining the physical trajectory of a particle between fixed spacetime endpoints. Stationarity need not imply a minimum.

7. Expectation-Value Functional in Quantum Mechanics

$$E[\psi] = \langle\psi|\widehat{H}|\psi\rangle$$

Assigns to each normalized wave function ψ its expected energy, provided it lies in the Hamiltonian's domain. Example: calculating the mean energy of an electron in hydrogen. For an unnormalized state, divide this expression by the state's squared norm.

8. Norm Functional in L^p

$$\| f \|_p = \left(\int_a^b | f(x)|^p \, dx \right)^{1/p}$$

Measures the "size" of a function in L^p, with the exponent at least one for this expression to be a norm. Example: comparing signal amplitudes in digital signal processing.

9. Curve-Length Functional

$$L[y] = \int_a^b \sqrt{1 + \left(y'(x)\right)^2} \, dx$$

Assigns to each curve $y(x)$ its length over $[a, b]$. Example: determining the distance traveled by an object in a plane.

10. Elastic Strain-Energy Functional

$$U[u] = \frac{1}{2} \int_\Omega \sigma(u) : \varepsilon(u) \, d\Omega$$

Here, σ is stress and ε is strain. In multidimensional linear elasticity, the product denotes double contraction of the stress and strain tensors. Example: calculating the energy stored in a deformed beam.

11. Least-Squares Error Functional

$$E[f] = \sum_{i=1}^{n} \left(y_i - f(x_i)\right)^2$$

Assigns to each function f the sum of squared residuals relative to experimental data. Example: fitting a line or polynomial to laboratory measurements.

12. Entropy Functional in Information Theory

$$H[p] = -\sum_i p_i \log p_i$$

Assigns to a discrete probability distribution p a measure of uncertainty. Example: analyzing entropy in statistical systems or digital communication. Probabilities are nonnegative and sum to one; the zero-probability contribution is defined by continuity. Natural logarithms give entropy in nats.

Examples of Functionals and Numerical Applications

Functional	**Mathematical Expression**	**Description**	**Application**	**Numerical Example**
Point evaluation	$F[f] = f(x_0)$	Evaluates a function at a fixed point x_0.	Interpolation and data analysis	If $f(x) = x^2$ and $x_0 = 3$, then $F[f] = 9$.

Functional	Mathematical Expression	Description	Application	Numerical Example
Simple integral	$F[f] = \int_a^b f(x)\,dx$	Calculates signed area under a curve.	Physics, engineering, and statistics	For $f(x) = x$ on $[0,2]$, $F[f] = 2$.
Time-integrated mechanical energy	$E[y] = \int_{t_1}^{t_2} \left(\frac{1}{2}m\dot{y}^2 + \frac{1}{2}ky^2\right) dt$	Integrates mechanical energy over time.	Classical mechanics	If $m = 1$, $k = 4$, and $y(t) = \cos t$ on $[0, \pi]$, then $E[y] = \frac{5\pi}{4}$.
Action	$S[y] = \int_{t_1}^{t_2} L(y, \dot{y}, t)\,dt$	Assigns an action to each trajectory.	Mechanics and field theory	For $L = \frac{1}{2}m\dot{y}^2 - \frac{1}{2}ky^2$ and $y(t) = \sin t$, calculate S on $[0, \pi]$.
Expectation value	$E[\psi] = \langle \psi \mid \widehat{H} \mid \psi \rangle$	Real for a self-adjoint Hamiltonian and a normalized state.	Quantum mechanics	Mean energy of a normalized wave function ψ.
Curve length	$L[y] = \int_a^b \sqrt{1 + (y'(x))^2}\,dx$	Calculates the length of a curve.	Differential geometry and CAD	For $y(x) = x^2$ on $[0,1]$, $L \approx 1.4789$.
Elastic strain energy	$U[u] = \frac{1}{2}\int_\Omega \sigma(u) : \varepsilon(u)\,d\Omega$	Measures energy stored by deformation.	Mechanical and civil engineering	For a uniaxial spring model: $U = \frac{1}{2}kx^2$, with $k = 200$, $x = 0.01$, and $U = 0.01$ J.
Least-squares error	$E[f] = \sum_{i=1}^{n} (y_i - f(x_i))^2$	Measures the squared fitting error.	Statistics and machine learning	For data points $(1,2)$ and $(2,3)$, with $f(x) = x$, $E = 2$.
Entropy	$H[p] = -\sum_i p_i \log p_i$	Measures uncertainty in a probability distribution.	Statistical physics and information theory	For $p = (0.5, 0.5)$, $H = \ln 2 \approx 0.6931$.

Functionals × Operators

Property	Functional	Operator
Input	Function	Function
Output	Number	Function
Example	$F[f] = \int_0^1 f(x)\,dx$	$T[f](x) = \frac{d}{dx}f(x)$

A functional can be regarded as a special type of operator whose output is a scalar rather than another function. Functionals are often constructed from operators: apply an operator to a function, then obtain a number through an inner product or integration. A quantum-mechanical example is

$$E[\psi] = \langle \psi \mid \widehat{H} \mid \psi \rangle$$

The Hamiltonian $\widehat{H}$ acts on the wave function ψ, and the inner product gives the expected energy for a normalized state.

2 Linear Functionals

A functional F on a vector space of functions X is linear if, for all $f, g \in X$ and scalars $\alpha, \beta \in \mathbb{R}$ (or $\mathbb{C}$),

$$F[\alpha f + \beta g] = \alpha F[f] + \beta F[g].$$

Thus, it preserves linear combinations. Examples:
Weighted Integral Functional

$$F[f] = \int_a^b f\,(x)\, w(x)\, dx$$

where $w(x)$ is fixed. This is linear because

$$F[\alpha f + \beta g] = \alpha \int_a^b f\,(x) w(x)\, dx + \beta \int_a^b g\,(x) w(x)\, dx.$$

Point-Evaluation Functional

$$F[f] = f(x_0).$$

This is linear because

$$F[\alpha f + \beta g] = \alpha f(x_0) + \beta g(x_0).$$

Inner Product with a Fixed Function. Let $f, g \in L^2[a, b]$. For real-valued functions,

$$F[f] = \langle f, g \rangle = \int_a^b f\,(x)\, g(x)\, dx$$

is linear in f. For complex functions, insert the complex conjugate of the fixed function in the integral. With the physicists' convention, this is the inner product with the fixed function in the bra and the variable function in the ket.
A Nonlinear Functional

$$G[f] = \int_a^b (f(x))^2 dx$$

This is generally nonlinear because

$$G[f + g] \neq G[f] + G[g].$$

A linear functional maps functions to numbers while preserving linear combinations. It is the function-space counterpart of a linear map in ordinary vector spaces. Linear functionals are

fundamental in functional analysis, differential equations, quantum mechanics, and approximation theory.

3 Functional Derivatives

Unless stated otherwise, the derivations below use real-valued fields and sufficiently smooth admissible variations. Complex orbitals and their conjugates are varied independently when indicated. All integrations are over the stated domain and required integrability and boundary conditions are assumed.

In differential calculus, a partial derivative measures how a function changes with one coordinate. For $f(x,y,z)$, the rate of change of f with respect to x is $\partial f/\partial x$. An analogous description is possible for a functional $I[u(x)]$. Here, $u(x)$ is the reference function; the direction of variation is an arbitrary admissible test function, not the reference function itself. The functional derivative $\delta I/\delta u$ of $I[u(x)]$ represents this directional response through an integral pairing.
Introduce a one-parameter family in $I[u(x)]$ and differentiate with respect to ϵ:

$$DI[u;\eta] = \frac{dI[u+\varepsilon\eta]}{d\varepsilon}\Big|_{\varepsilon=0} = lim_{\varepsilon\to 0}\frac{I[u+\varepsilon\eta]-I[u]}{\varepsilon}$$
$$= lim_{\varepsilon\to 0}\frac{1}{\varepsilon}\int_{x_0}^{x_1}[F(x,u+\varepsilon\eta,u'+\varepsilon\eta')-F(x,u,u')]\,dx\,. \tag{1}$$

Assuming sufficient smoothness to differentiate under the integral sign, a multivariable Taylor expansion of the first integrand gives

$$F(x,u+\varepsilon\eta,u'+\varepsilon\eta') = F(x,u,u') + \varepsilon\frac{\partial F}{\partial u}\eta + \varepsilon\frac{\partial F}{\partial u'}\eta' + O(\varepsilon^2),$$
$$DI[u;\eta] = lim_{\varepsilon\to 0}\frac{1}{\varepsilon}\int_{x_0}^{x_1}\left[\varepsilon\frac{\partial F}{\partial u}\eta + \varepsilon\frac{\partial F}{\partial u'}\eta' + O(\varepsilon^2)\right]dx\,.$$

After division by the variation parameter, the contribution from terms of order $O(\epsilon^2)$ vanishes in the limit. Canceling the unperturbed integrals gives

$$DI[u;\eta] = lim_{\varepsilon\to 0}\int_{x_0}^{x_1}\left(\frac{\partial F}{\partial u}\eta + \frac{\partial F}{\partial u'}\eta' + O(\varepsilon)\right)dx$$
$$= \int_{x_0}^{x_1}\left(\frac{\partial F}{\partial u}\eta + \frac{\partial F}{\partial u'}\eta'\right)dx\,.$$
$$\frac{dI}{d\epsilon}\Big|_{\epsilon=0} = \int_{x_0}^{x_1}\left(\frac{\partial F}{\partial u}\,\eta + \frac{\partial F}{\partial u'}\,\eta'\right)dx.$$

Integrating the derivative term by parts gives

$$\left.\frac{dI}{d\epsilon}\right|_{\epsilon=0} = \int_{x_0}^{x_1} \frac{\partial F}{\partial u}\eta(x)dx + \underbrace{\left[\frac{\partial F}{\partial u'}\eta(x)\right]_{x_0}^{x_1}}_{=0} - \int_{x_0}^{x_1} \frac{d}{dx}\left(\frac{\partial F}{\partial u'}\right)\eta(x)dx.$$

The boundary term vanishes because $\eta(x_0) = \eta(x_1) = 0$. Hence,

$$\left.\frac{dI}{d\epsilon}\right|_{\epsilon=0} = \int_{x_0}^{x_1} \left[\frac{\partial F}{\partial u} - \frac{d}{dx}\left(\frac{\partial F}{\partial u'}\right)\right]\eta(x)dx = \int_{x_0}^{x_1} \frac{\delta I}{\delta u}\eta(x)dx.$$

This identifies the functional derivative as

$$\frac{\delta I}{\delta u} = \frac{\partial F}{\partial u} - \frac{d}{dx}\left(\frac{\partial F}{\partial u'}\right).$$

The first directional derivative of $I[u, u']$ admits a functional-derivative representation when it can be written as

$$\left.\frac{dI}{d\epsilon}\right|_{\epsilon=0} = \int_{x_0}^{x_1} \frac{\delta I}{\delta u}\eta(x)dx, \tag{2}$$

The result must be expressed as the pairing on the right-hand side of Equation (2), rather than identifying the directional derivative with its coefficient pointwise (Parr and Yang, 1989, p. 246). In more general spaces, the derivative may be a distribution rather than an ordinary function.

Example 1. Using Equation (2), find the derivative of

$$F[y] = \int_{x_0}^{x_1} y(x)^2 dx\,.$$

$$DF[y;\eta] = \lim_{\epsilon\to 0}\frac{1}{\epsilon}\int_{x_0}^{x_1}[(y+\epsilon\eta)^2 - y^2]dx = \lim_{\epsilon\to 0}\frac{1}{\epsilon}\int_{x_0}^{x_1}(y^2 + 2y\epsilon\eta + \epsilon^2\eta^2 - y^2)dx$$

$$= lim_{\varepsilon\to 0}\int_{x_0}^{x_1}(2y\eta + \varepsilon\eta^2)dx$$

$$= \int_{x_0}^{x_1} 2y(x)\eta(x)dx\,.$$

The result has the form of Equation (2). Therefore, the functional derivative of $F[y(x)]$ is

$$\frac{\delta F}{\delta y(x)} = 2y(x).$$

Example 2. Find the functional derivative of

$$F[\varphi] = \int_{t_0}^{t_1} \sqrt{\varphi(t)}\, dt.$$

Assume the reference function is positive and bounded away from zero on the integration interval, so sufficiently small admissible perturbations remain positive. Expanding $\left(\varphi(t) + \varepsilon\phi(t)\right)^{\frac{1}{2}}$ in a binomial series gives

$$\sqrt{\varphi+\varepsilon\phi}=\sqrt{\varphi}+\frac{\varepsilon\phi}{2\sqrt{\varphi}}-\frac{\varepsilon^2\phi^2}{8\varphi^{\frac{3}{2}}}+O(\varepsilon^3).$$

Using the definition, we have

$$\begin{aligned}DF[\varphi;\phi]=\left.\frac{dF[\varphi;\phi]}{d\epsilon}\right|_{\epsilon=0}&=\lim_{\epsilon\to0}\frac{1}{\epsilon}\int_{t_0}^{t_1}\left[\left(\varphi(t)+\varepsilon\phi(t)\right)^{\frac{1}{2}}-\varphi^{\frac{1}{2}}(t)\right]\\&=\lim_{\epsilon\to0}\frac{1}{\epsilon}\int_{x_0}^{x_1}\left[\varphi^{\frac{1}{2}}(t)+\frac{1}{2}\varphi^{-\frac{1}{2}}(t)\varepsilon\phi(t)+O(\varepsilon^2)-\varphi^{\frac{1}{2}}(t)\right]dt\\&=\lim_{\epsilon\to0}\frac{1}{\epsilon}\int_{x_0}^{x_1}\left[\frac{1}{2}\varphi^{-\frac{1}{2}}(t)\varepsilon\phi(t)+O(\varepsilon^2)\right]dt=\int\frac{1}{2}\varphi^{-\frac{1}{2}}(t)\phi(x)dt\end{aligned}$$

This has the form of Equation (2), so the functional derivative of $F[\varphi(t)]$ is

$$\frac{\delta F[\varphi(t)]}{\delta\varphi}=\frac{1}{2}\varphi^{-\frac{1}{2}}(t).$$

4 Variational Notation δ

We use δ to take advantage of the analogy with the differential symbols d and ∂, without writing the parameter ϵ and direction η at every step. Ordinary differentials such as d and ∂ describe changes associated with moving between points. By contrast, δ compares neighboring functions: in particular, δ denotes a change of function at fixed x. The symbols d and ∂ retain their usual differential meanings.

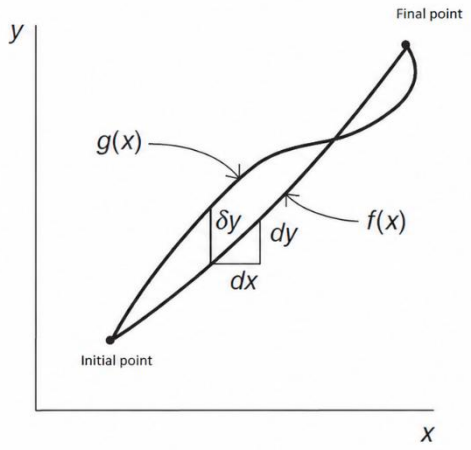


Figure 1. Ordinary and variational changes. δ compares neighboring functions at fixed independent variable, whereas d describes change along a function. Thus, δ denotes an infinitesimal change of $f(x)$ at fixed x, while d denotes the change in f caused by an infinitesimal change of x.

To examine the analogy between δ and d, consider

$$I=\int_{x_0}^{x_1}F(x,u,u')dx.$$

Consider functions with fixed endpoint values at x_0 and x_1. We seek $u(x)$ that makes $I\left[u(x), u'^{(x)}\right]$ stationary:

$$\delta\, I[u] = \delta \int_{x_0}^{x_1} F(x, u, u')dx = \int_{x_0}^{x_1} \delta\, F dx = 0. \tag{3}$$

Equation (3) uses the fact that variation commutes with integration over fixed limits. With δ, compare the ordinary differential of $f(x, y, z)$,

$$df = \frac{\partial f}{\partial x}dx + \frac{\partial f}{\partial y}dy + \frac{\partial f}{\partial z}dz$$

which describes the change of f between points, with the differential of $F\big(x, u(x), u'(x)\big)$:

$$dF = \frac{\partial F}{\partial x}dx + \frac{\partial F}{\partial u}du + \frac{\partial F}{\partial u'}du'.$$

The variation is distinct from the ordinary differential: df and dF describe pointwise differentials, while replacing dF by its variational counterpart gives

$$\delta F = \frac{\partial F}{\partial x}\delta x + \frac{\partial F}{\partial u}\delta u + \frac{\partial F}{\partial u'}\delta u'. \tag{4}$$

For variations at fixed independent variable, x does not change, so $\delta x = 0$. Therefore,

$$\delta F = \frac{\partial F}{\partial u}\delta u + \frac{\partial F}{\partial u'}\delta u'$$

This describes the change of F between neighboring functions. In ordinary calculus, stationarity of $f(x, y, z)$ requires $df = 0$ for every direction. Similarly, stationarity of a functional requires $\delta I = 0$ for every admissible variation of $u(x)$, with I denoting the functional. The sum and product rules for df also apply to δF:

$$\delta(F + G) = \delta F + \delta G \text{ and } \delta(FG) = F\delta G + G\delta F.$$

To illustrate δ, derive the Euler–Lagrange equation again by seeking a function that makes

$$I[u] = \int_{x_0}^{x_1} F[x, u(x), u'(x)]dx$$

stationary. Requiring $\delta I = 0$ gives

$$\delta I[u] = \delta \int_{x_0}^{x_1} F[x, u(x), u'(x)]dx = \int_{x_0}^{x_1} \delta F[x, u(x), u'(x)]dx = 0.$$

Using Equation (4),

$$\begin{aligned}\delta\, I &= \int_{x_0}^{x_1} \left(\frac{\partial\, F}{\partial\, x}\delta\, x + \frac{\partial\, F}{\partial\, u}\delta\, u + \frac{\partial\, F}{\partial\, u'}\delta\, u'\right) dx \\ &= \int_{x_0}^{x_1} \left(\frac{\partial\, F}{\partial\, u}\delta\, u + \frac{\partial\, F}{\partial\, u'}\delta\, u'\right) dx\,.\end{aligned}$$

Here, x is fixed, so $\delta x = 0$. Integrating the derivative term by parts gives

$$\delta I = \int_{x_0}^{x_1} \frac{\partial F}{\partial u}\delta u dx + \left[\frac{\partial F}{\partial u'}\delta u\right]_{x_0}^{x_1} - \int_{x_0}^{x_1} \frac{d}{dx}\left(\frac{\partial F}{\partial u'}\right)\delta u dx.$$

Since $u(x)$ has fixed endpoint values,

$$\delta I = \int_{x_0}^{x_1} \frac{\partial F}{\partial u}\delta u dx + \underbrace{\left[\frac{\partial F}{\partial u'}\delta u\right]_{x_0}^{x_1}}_{=0} - \int_{x_0}^{x_1} \frac{d}{dx}\left(\frac{\partial F}{\partial u'}\right)\delta u dx = \int_{x_0}^{x_1} \left[\frac{\partial F}{\partial u} - \frac{d}{dx}\left(\frac{\partial F}{\partial u'}\right)\right]\delta u dx = 0.$$

The fundamental lemma of the calculus of variations then gives

$$\frac{\partial F}{\partial u} - \frac{d}{dx}\left(\frac{\partial F}{\partial u'}\right) = 0$$

This is exactly the Euler–Lagrange equation for $F = F(x, u, u')$ obtained previously.

5 Sum Rule

Equation (1) yields rules analogous to those of ordinary differentiation. First consider the sum rule, or linearity of differentiation, for functionals $F[f(x)]$ and $G[f(x)]$ and constants c_1 and c_2:

$$\frac{\delta}{\delta f(x)}(c_1 F[f(x)] + c_2 G[f(x)]) = c_1 \frac{\delta F[f(x)]}{\delta f(x)} + c_2 \frac{\delta G[f(x)]}{\delta f(x)}.$$

Proof

Let $I[f(x)] = c_1 F[f(x)] + c_2 G[f(x)]$

$$\begin{aligned}
\frac{\delta I[f(x)]}{\delta f(x)} = \frac{dI}{d\epsilon}\Big|_{\epsilon=0} &= \lim_{\epsilon\to 0} \frac{I[f+\epsilon\eta] - I[f]}{\epsilon} \\
&= \lim_{\epsilon\to 0} \frac{c_1 F[f(x)+\varepsilon\phi(x)] + c_2 G[f(x)+\varepsilon\phi(x)] - c_1 F[f(x)] - c_2 G[f(x)]}{\epsilon} \\
&= \lim_{\epsilon\to 0} \frac{c_1\{F[f(x)+\varepsilon\phi(x)] - F[f(x)]\} + c_2\{G[f(x)+\varepsilon\phi(x) - G[f(x)]]\}}{\epsilon} \\
&= \lim_{\epsilon\to 0} \left\{\frac{c_1\{F[f(x)+\varepsilon\phi(x)] - F[f(x)]\}}{\epsilon} + \frac{c_2\{G[f(x)+\varepsilon\phi(x)] - G[f(x)]\}}{\epsilon}\right\} \\
&= c_1 \lim_{\epsilon\to 0} \frac{F[f(x)+\varepsilon\phi(x)] - F[f(x)]}{\epsilon} + c_2 \lim_{\epsilon\to 0} \frac{G[f(x)+\varepsilon\phi(x)] - G[f(x)]}{\epsilon} \\
&= c_1 \frac{\delta F[f(x)]}{\delta f(x)} + c_2 \frac{\delta G[f(x)]}{\delta f(x)}
\end{aligned}$$

Because the test function is arbitrary, equality of the directional derivatives implies the stated equality of functional derivatives. Thus, differentiation commutes with constant linear combinations.

6 Product Rule

Consider the product of $F[f(x)]$ and $G[f(x)]$:

$$I[f(x)] = F[f(x)]G[f(x)]$$

Its functional derivative is

$$\boxed{\frac{\delta}{\delta f(x)}(FG) = \frac{\delta F}{\delta f(x)}G + F\frac{\delta G}{\delta f(x)}}$$

Proof

$$\frac{\delta I[f(x)]}{\delta f(x)} = \frac{dI}{d\epsilon}\Big|_{\epsilon=0} = \lim_{\epsilon\to 0}\frac{I[u+\epsilon\eta]-I[u]}{\epsilon} =$$

$$= \lim_{\epsilon\to 0}\frac{F[f(x)+\varepsilon\phi(x)]G[f(x)+\varepsilon\phi(x)] - F[f(x)]G[f(x)]}{\epsilon}$$

$$= \lim_{\epsilon\to 0}\frac{F[f+\varepsilon\phi]G[f+\varepsilon\phi] + F[f+\varepsilon\phi]G[f] - F[f+\varepsilon\phi]G[f] - F[f]G[f]}{\epsilon}$$

$$= \lim_{\epsilon\to 0}\left\{\frac{F[f(x)+\varepsilon\phi(x)]\{G[f(x)+\varepsilon\phi(x)] - G[f(x)]\}}{\epsilon}\right.$$

$$\left. + \frac{\{F[f(x)+\varepsilon\phi(x) - F[f(x)]\}G[f(x)]}{\epsilon}\right\}$$

$$= \lim_{\epsilon\to 0} F[f(x)+\varepsilon\phi(x)] \lim_{\epsilon\to 0}\frac{G[f(x)+\varepsilon\phi(x)] - G[f(x)]}{\epsilon}$$

$$+ \lim_{\epsilon\to 0}\frac{F[f(x)+\varepsilon\phi(x)] - F[f(x)}{\epsilon} \lim_{\epsilon\to 0} G[f(x)+\varepsilon\phi(x)] = F\frac{\delta G}{\delta f(x)} + \frac{\delta F}{\delta f(x)}G$$

7 Chain Rule

Let F be a functional of $f(x)$, written $F = F[f(x)]$. Its first-order change δF is

$$\delta F = \int \frac{\delta F}{\delta f(x)}\delta f(x)dx. \tag{5}$$

Now suppose f, at each x, is itself a functional of $g(x)$, denoted by $f = f[x, g(x)]$. Its variation δf is

$$\delta f = \int \frac{\delta f}{\delta g(x')}\delta g(x')dx'. \tag{6}$$

Substituting Equation (6) into Equation (5), and interchanging integrations when justified, gives

$$\begin{aligned}\delta F &= \int\int \frac{\delta F}{\delta f(x)}\frac{\delta f(x)}{\delta g(x')}\delta g(x')\,dx'\,dx \\ &= \int\left[\int \frac{\delta F}{\delta f(x)}\frac{\delta f(x)}{\delta g(x')}\,dx\right]\delta g(x')\,dx'.\end{aligned} \tag{7}$$

Collecting the coefficient of the arbitrary variation in Equation (7), as in Equation (2), yields the functional chain rule:

$$\frac{\delta F}{\delta g(x')} = \int \frac{\delta F}{\delta f(x)}\frac{\delta f(x)}{\delta g(x')}dx. \tag{8}$$

For a local composition in which f is an ordinary differentiable function of the value of the other field at the same point, the kernel contains a Dirac delta. The integral therefore reduces to

$$\frac{\delta F}{\delta g(x)} = \frac{\delta F}{\delta f(x)} \frac{df}{dg(x)} \tag{9}$$

Here, $F = F\big[f\big(g(x)\big)\big]$. To establish Equation (9), begin with $F = F[f(x)]$. Its change δF due to a change of $f(x)$ is, for the functional F,

$$\delta F = \int \frac{\delta F}{\delta f(x)} \delta f(x) dx. \tag{10}$$

If f is a local function of $g(x)$, so that $f = f\big(g(x)\big)$, its variation f at fixed x, induced by $g(x)$, is

$$\delta f = \frac{\partial f}{\partial g} \delta g \tag{11}$$

Equation (11) uses the fixed-coordinate variation in Equation (4). Substituting Equation (11) into Equation (10) gives

$$\delta F = \int \frac{\delta F}{\delta f} \frac{\partial f}{\partial g} \delta g \, dx. \tag{12}$$

Equation (12) has the form of Equation (2), so

$$\frac{\delta F}{\delta g} = \frac{\delta F}{\delta f} \frac{\partial f}{\partial g} \tag{13}$$

As an application of Equation (13), consider the energy functional

$$E = E[\rho(\boldsymbol{r})].$$

For a closed-shell system with doubly occupied spatial orbitals, the electron density ρ is

$$\rho(\boldsymbol{r}) = 2\sum_{i}^{N/2} |\phi_i(\boldsymbol{r})|^2 = 2\sum_{i}^{N/2} \phi_i^*(\boldsymbol{r})\phi_i(\boldsymbol{r}).$$

Varying the orbital and its complex conjugate independently, the energy derivative with respect to ϕ_j^* is

$$\frac{\delta \rho(r')}{\delta \varphi_j^*(r)} = 2\varphi_j(r')\delta(r' - r),$$

$$\frac{\delta\, E}{\delta \varphi_j^*(r)} = \int \frac{\delta\, E}{\delta \rho(r')} 2\varphi_j(r')\delta(r' - r)\, dr'$$

$$= 2\frac{\delta\, E}{\delta \rho(r)} \varphi_j(r).$$

Example 2. Consider

$$F[g(x)] = \int g(x)^2 dx,$$

where

$$g(x) = \int k(x, y) f(y) dy$$

and $k(x,y)$ is a prescribed kernel. We seek the derivative of F with respect to $f(x)$, denoted by $\delta F/\delta f(x)$. The chain rule gives

$$\frac{\delta F}{\delta f(x)} = \int \frac{\delta F}{\delta g(y)} \frac{\delta g(y)}{\delta f(x)} dy$$

a) First, calculate $\frac{\delta F}{\delta g(y)}$:

Since $F[g(x)] = \int g(x)^2 dx$,

we obtain

$$\frac{\delta F}{\delta g(y)} = 2g(y).$$

b) Next, calculate $\frac{\delta g(y)}{\delta f(x)}$:

Since

$$g(x) = \int k(x,y) f(y) dy$$

we obtain

$$\frac{\delta\, g(y)}{\delta\, f(x)} = k(y,x).$$

c) Substitution into the chain rule gives the following result. The argument order of the kernel is essential; no symmetry of the kernel is assumed.

$$\frac{\delta\, F}{\delta\, f(x)} = 2 \int g(y) k(y,x) dy$$

$$= 2 \int k(y,x) \left[\int k(y,z) f(z)\, dz \right] dy.$$

Example 3. Let $F[g]$ be defined by

$$F[g] = \int e^{g(x)} dx$$

with

$$g(x) = \int f(y) h(x-y) dy$$

where $h(x-y)$ is prescribed. We seek the derivative of F with respect to $f(x)$:

$$\frac{\delta F}{\delta f(x)}$$

First calculate $\frac{\delta F}{\delta g(y)}$:

Since

$$F[g] = \int e^{g(x)} dx\,,$$

we obtain

$$\frac{\delta F}{\delta g(y)} = e^{g(y)}$$

Next, calculate

$$\frac{\delta g(y)}{\delta f(x)}:$$

Since

$$g(x) = \int f(y)h(x-y)dy,$$

we obtain

$$\frac{\delta\, g(y)}{\delta\, f(x)} = h(y-x).$$

Substitution into the chain rule gives

$$\frac{\delta\, F}{\delta\, f(x)} = \int e^{g(y)}h(y-x)\, dy$$

$$= \int exp\left[\int f(z)h(y-z)\, dz\right]h(y-x)dy\,.$$

8 Functional Derivatives in One Dimension

As shown above, the derivative of

$$F[y(x)] = \int f\big(x, y(x), \mathrm{y}'(x)\big)dx.$$

is

$$\frac{\delta F[y]}{\delta y} = \frac{\partial f}{\partial y} - \frac{d}{dx}\left(\frac{\partial f}{\partial y'}\right). \tag{14}$$

This formula generalizes to

$$F[y(x)] = \int f\big(x, y, \mathrm{y}^{(1)}, \mathrm{y}^{(2)}, \mathrm{y}^{(3)}, \cdots\big)dx,$$

where $y = y(x)$, and $\mathrm{y}^{(i)}$ denotes the derivative of order i of y with respect to x. The general expression is

$$\frac{\delta F[y]}{\delta y} = \frac{\partial f}{\partial y} - \frac{d}{dx}\left(\frac{\partial f}{\partial y^{(1)}}\right) + \frac{d^2}{dx^2}\left(\frac{\partial f}{\partial y^{(2)}}\right) - \frac{d^3}{dx^3}\left(\frac{\partial f}{\partial y^{(3)}}\right) + \cdots$$

$$\frac{\delta F[y]}{\delta y} = \sum_{i=0}^{n}(-1)^i \frac{d^i}{dx^i}\left(\frac{\partial f}{\partial y^{(i)}}\right). \tag{15}$$

To derive Equation (15), apply the same variation as for Equation (14), then integrate the term containing each derivative of the variation by parts as many times as its order. Each integration contributes a minus sign. Assume the variation and its derivatives up to one less than the highest order vanish at both endpoints; otherwise, retain the boundary terms to obtain the appropriate natural boundary conditions.

9 Functional Derivatives in Several Dimensions

Consider

$$F[\rho] = \int f\big(\boldsymbol{r}, \rho(\boldsymbol{r}), \nabla\rho(\boldsymbol{r})\big)d\boldsymbol{r}. \tag{16}$$

Let $\boldsymbol{\phi}(\boldsymbol{r})$ vanish on the boundary of the integration domain. Define $Y(\boldsymbol{r}, \epsilon)$ by

$$Y(\boldsymbol{r}, \epsilon) = \rho(\boldsymbol{r}) + \epsilon\boldsymbol{\phi}(\boldsymbol{r}) \quad \text{and} \quad \nabla Y(\boldsymbol{r}, \epsilon) = \nabla\rho(\boldsymbol{r}) + \epsilon\nabla\boldsymbol{\phi}(\boldsymbol{r}).$$

The functional derivative of Equation (16) is

$$\begin{aligned}
\left.\frac{dF}{d\epsilon}\right|_{\epsilon=0} &= \left[\frac{d}{d\epsilon}\int f\big(\boldsymbol{r}, Y(\boldsymbol{r}), \nabla Y(\boldsymbol{r})\big)d\boldsymbol{r}\right]_{\epsilon=0} \\
&= \left[\int \frac{\partial f}{\partial Y}\frac{\partial Y}{\partial \epsilon} + \frac{\partial f}{\partial(\nabla Y)}\frac{\partial(\nabla Y)}{\partial \epsilon}\,d\boldsymbol{r}\right]_{\epsilon=0} \\
&= \int \left(\frac{\partial f}{\partial \rho}\phi + \frac{\partial f}{\partial \nabla\rho}\cdot\nabla\phi\right)d\boldsymbol{r} \\
&= \int \left[\frac{\partial f}{\partial \rho}\phi + \nabla\cdot\left(\frac{\partial f}{\partial \nabla\rho}\phi\right) - \left(\nabla\cdot\frac{\partial f}{\partial \nabla\rho}\right)\phi\right]d\boldsymbol{r} \\
&= \int \left[\frac{\partial f}{\partial \rho}\phi - \left(\nabla\cdot\frac{\partial f}{\partial \nabla\rho}\right)\phi\right]d\boldsymbol{r} \\
&= \int \left(\frac{\partial f}{\partial \rho} - \nabla\cdot\frac{\partial f}{\partial \nabla\rho}\right)\phi(\boldsymbol{r})d\boldsymbol{r}
\end{aligned}$$

$$DF[\rho;\varphi] = \int \left(\frac{\partial f}{\partial \rho} - \nabla\cdot\frac{\partial f}{\partial \nabla\rho}\right)\phi(\boldsymbol{r})d\boldsymbol{r} \tag{17}$$

Evaluate the partial derivatives at $\epsilon = 0$. The vector derivative $\partial f/\partial\nabla\rho$ is defined by

$$\frac{\partial f}{\partial \nabla\rho} \equiv \frac{\partial f}{\partial \rho_x}\hat{\imath} + \frac{\partial f}{\partial \rho_y}\hat{\jmath} + \frac{\partial f}{\partial \rho_z}\hat{k},$$

Here,

$$\rho_x = \frac{\partial \rho}{\partial\, x}, \qquad \rho_y = \frac{\partial \rho}{\partial\, y}, \qquad \rho_z = \frac{\partial \rho}{\partial\, z}$$

The next step uses the divergence product rule:

$$\nabla\cdot\left(\frac{\partial f}{\partial \nabla\rho}\phi\right) = \left(\nabla\cdot\frac{\partial f}{\partial \nabla\rho}\right)\phi + \frac{\partial f}{\partial \nabla\rho}\cdot\nabla\phi.$$

The divergence term becomes a surface integral by Gauss's theorem and vanishes because $\boldsymbol{\phi}(\boldsymbol{r}) = 0$ on the boundary:

$$\int \nabla\cdot\left(\frac{\partial f}{\partial \nabla\rho}\phi(\boldsymbol{r})\right)d\boldsymbol{r} = \oiint\left(\frac{\partial f}{\partial \nabla\rho}\phi(\boldsymbol{r})\right)\cdot d\boldsymbol{s} = 0,$$

Here, $d\boldsymbol{s}$ is the outward-oriented surface-area vector, not merely a unit normal, and S is the closed boundary surface. The surface term vanishes because $\phi(\boldsymbol{r}) = 0$ there. Equation (17), therefore, has the form

$$\left.\frac{dF}{d\epsilon}\right|_{\epsilon=0} = \int \frac{\delta F}{\delta\rho(\boldsymbol{r})}\phi(\boldsymbol{r})d\boldsymbol{r}$$

The functional derivative of $F[\rho(\boldsymbol{r})]$ is consequently

$$\boxed{\frac{\delta F}{\delta\rho(\boldsymbol{r})} = \frac{\partial f}{\partial\rho} - \nabla\cdot\frac{\partial f}{\partial\nabla\rho}} \tag{18}$$

Here, $\rho = \rho(\boldsymbol{r})$ and $f = f\big(\boldsymbol{r}, \rho(\boldsymbol{r}), \nabla\rho(\boldsymbol{r})\big)$. Equation (18) applies to functionals of the form

$$F[\rho(\boldsymbol{r})] = \int f\big(\boldsymbol{r}, \rho(\boldsymbol{r}), \nabla\rho(\boldsymbol{r})\big)d\boldsymbol{r}.$$

It generalizes to functionals involving higher spatial derivatives:

$$F[\rho(\boldsymbol{r})] = \int f\left(\boldsymbol{r}, \rho(\boldsymbol{r}), \nabla\rho(\boldsymbol{r}), \nabla^{(2)}\rho(\boldsymbol{r}), \nabla^{(3)}\rho(\boldsymbol{r}), \cdots, \nabla^{(N)}\rho(\boldsymbol{r})\right)d\boldsymbol{r}, \tag{19}$$

Here, $\boldsymbol{r} \in \mathbb{R}^n$, and $\nabla^{(i)}$ denotes the tensor of differential operators of order i, with n^i ordered components. In this section, powers of the gradient denote full derivative tensors; in particular, the second-order tensor is the Hessian operator, not the scalar Laplacian used in the kinetic-energy sections.

$$\left[\nabla^{(i)}\right]_{\alpha_1\alpha_2\cdots\alpha_i} = \frac{\partial^i}{\partial r_{\alpha_1}\partial r_{\alpha_2}\cdots\partial r_{\alpha_i}}$$

The indices $\alpha_1, \alpha_2, \cdots, \alpha_i = 1,2,\cdots,n$ label Cartesian coordinate directions. For $n = 3$ spatial dimensions and derivative order $i = 1$, the components are

$$[\nabla]_\alpha = \frac{\partial}{\partial r_\alpha} \Longrightarrow \left(\frac{\partial}{\partial r_1}, \frac{\partial}{\partial r_2}, \frac{\partial}{\partial r_3}\right) = \left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z}\right),$$

For clarity, the last equality uses $r_1 = x, r_2 = y$ and $r_3 = z$. In $(n = 3)$ dimensions and at derivative order $(i = 2)$, the components are

$$\left[\nabla^{(2)}\right]_{\alpha\beta} = \frac{\partial^2}{\partial r_\alpha \partial r_\beta},$$

With $\alpha, \beta = 1,2,3$ and coordinates x, y, z, the matrix representation is

$$\begin{pmatrix} \frac{\partial^2}{\partial x^2} & \frac{\partial^2}{\partial x\partial y} & \frac{\partial^2}{\partial x\partial z} \\ \frac{\partial^2}{\partial y\partial x} & \frac{\partial^2}{\partial y^2} & \frac{\partial^2}{\partial y\partial z} \\ \frac{\partial^2}{\partial z\partial x} & \frac{\partial^2}{\partial z\partial y} & \frac{\partial^2}{\partial z^2} \end{pmatrix}$$

For Equation (19), repeated integration by parts gives the following generalization of Equation (18). Assume the variation and the normal-derivative traces required to eliminate the boundary terms vanish, or use compactly supported variations to obtain the interior derivative:

$$\frac{\delta F[\rho]}{\delta \rho} = \frac{\partial f}{\partial \rho} - \nabla \cdot \frac{\partial f}{\partial (\nabla \rho)} + \nabla^{(2)} \cdot \frac{\partial f}{\partial (\nabla^{(2)} \rho)} - \cdots + (-1)^N \nabla^{(N)} \cdot \frac{\partial f}{\partial (\nabla^{(N)} \rho)}$$

$$\boxed{\frac{\delta F[\rho]}{\delta \rho} = \frac{\partial f}{\partial \rho} + \sum_{i=1}^{N} (-1)^i \nabla^{(i)} \cdot \frac{\partial f}{\partial (\nabla^{(i)} \rho)}} \tag{20}$$

The n^i components of $\partial f / \partial \left(\nabla^{(i)} \rho\right)$ are the partial derivatives of f with respect to the corresponding derivatives of ρ:

$$\left[\frac{\partial f}{\partial (\nabla^{(i)} \rho)}\right]_{\alpha 1 \alpha 2 \cdots \alpha i} = \frac{\partial f}{\partial \rho_{\alpha 1 \alpha 2 \cdots \alpha i}}$$

where

$$\rho_{\alpha 1 \alpha 2 \cdots \alpha i} = \frac{\partial^i \rho}{\partial r_{\alpha 1} \partial r_{\alpha 2} \cdots \partial r_{\alpha i}}.$$

The contraction of the differential-operator tensor with the coefficient tensor is

$$\nabla^{(i)} \cdot \frac{\partial f}{\partial (\nabla^{(i)} \rho)} = \sum_{\alpha 1, \alpha 2, \cdots, \alpha i = 1}^{n} \frac{\partial^i}{\partial r_{\alpha 1} \partial r_{\alpha 2} \cdots \partial r_{\alpha i}} \frac{\partial f}{\partial \rho_{\alpha 1 \alpha 2 \cdots \alpha i}}.$$

For example, with $n = 3$ and $i = 2$, the contraction is

$$\nabla^2 \cdot \frac{\partial f}{\partial (\nabla^2 \rho)} = \sum_{\alpha=1}^{3} \sum_{\beta=1}^{3} \frac{\partial^2}{\partial r_\alpha \partial r_\beta} \left(\frac{\partial f}{\partial \rho_{\alpha \beta}} \right).$$

where

$$\rho_{\alpha \beta} = \frac{\partial^2 \rho}{\partial r_\alpha \partial r_\beta}.$$

An alternative notation is

$$\mathcal{S}[u] = \int_\Omega \mathcal{L}\left(u,\ \partial u,\ \partial^2 u, \ldots, \partial^N u,\ \mathbf{r}\right) d\mathbf{r}$$

Its terms have the following meanings. For clarity, the formulas below are for a scalar field; vector fields require one equation per component. $u = u(\mathbf{r})$: the unknown field on $\Omega \subset \mathbb{R}^d$. ∂u: all first partial derivatives, or the gradient. In two dimensions, $\partial u = \left(u_x, u_y\right)$; in three dimensions, $\partial u = \left(u_x, u_y, u_z\right)$. $\partial^2 u$: the second partial derivatives, or the Hessian. In two dimensions, $\partial^2 u = \left(u_{xx}, u_{xy}, u_{yx}, u_{yy}\right)$, with $u_{xy} = u_{yx}$ for $u \in C^2$. In three dimensions, the entries are $u_{x_i x_j}$, with

$i, j \in 1,2,3$. $\partial^k u$: all partial derivatives of order k. In multi-index notation, $D^\alpha u = \partial^{|\alpha|} u / \partial x_1^{\alpha_1} \cdots \partial x_d^{\alpha_d}$, with $|\alpha| = k$. $\mathbf{r}$: explicit spatial position; $\mathcal{L}$ may depend on $\mathbf{r}$. $d\mathbf{r}$: the volume element, namely dx in one dimension, $dx\,dy$ in two, or $dx\,dy\,dz$ in three.

An explicit two-dimensional form through second order, listing each distinct mixed derivative once, is

$$\mathcal{S}[u] = \int_\Omega \mathcal{L}\left(u,\ u_x, u_y,\ u_{xx}, u_{xy}, u_{yy},\ x, y\right) dx\,dy.$$

In three dimensions, through order N,

$$\mathcal{S}[u] = \int_\Omega \mathcal{L}\left(u,\ u_x, u_y, u_z,\ u_{xx}, u_{xy}, \ldots, u_{zz},\ \ldots,\ \{D^\alpha u\}_{|\alpha|=N},\ x, y, z\right) dx\,dy\,dz.$$

For example, in three dimensions with $\mathcal{L} = 1/2\,|\nabla u|^2 + V(u)$,

$$\mathcal{S}[u] = \int_\Omega \left[1/2\left(u_x^2 + u_y^2 + u_z^2\right) + V(u)\right] dV.$$

The interior stationarity condition for $\mathcal{S}$, through derivative order N, is

$$\sum_{|\alpha|=0}^{N} (-1)^{|\alpha|} D^\alpha \left(\frac{\partial\, L}{\partial (D^\alpha u)}\right) = 0.$$

Here each distinct multi-index is counted once. This convention avoids double counting mixed derivatives; the earlier full-tensor notation sums ordered components and must be used with a consistent symmetric-tensor derivative convention. Boundary terms determine the natural boundary conditions when the corresponding traces are not prescribed.

10 The Dirac Delta Function

Point evaluation can itself be expressed as a functional using the Dirac delta distribution. For a function $\rho(\boldsymbol{r})$ of $\boldsymbol{r}$,

$$\rho(\boldsymbol{r}) = F[\rho] = \int \rho(\boldsymbol{r}')\delta(\boldsymbol{r} - \boldsymbol{r}')d\boldsymbol{r}'.$$

The functional derivative of this evaluation functional is the identity kernel:

$$\frac{\delta\rho(\boldsymbol{r})}{\delta\rho(\boldsymbol{r}')} \equiv \frac{\delta F[\rho]}{\delta\rho(\boldsymbol{r}')} = \frac{\partial}{\partial\rho(\boldsymbol{r}')}[\rho(\boldsymbol{r}')\delta(\boldsymbol{r} - \boldsymbol{r}')] = \delta(\boldsymbol{r} - \boldsymbol{r}').$$

The Dirac delta, denoted by $\delta(x)$, is a distribution rather than an ordinary function. It is a continuous linear functional on smooth, compactly supported test functions. Introduced into physics by Paul Dirac, it represents quantities concentrated at a point, such as point charges or localized particles. A useful heuristic is that it has unit mass concentrated at the origin:

$$\int_{-\infty}^{\infty} \delta(x)\, dx = 1, \qquad \operatorname{supp} \delta = \{0\}.$$

The delta has no finite or infinite pointwise value that defines it as an ordinary function. Its rigorous meaning is given by its action on test functions. It can also be regarded as the unit point-mass probability measure.

Formally, its action on a test function $\phi(x)$ is

$$\int_{-\infty}^{\infty} \delta(x-a)\phi(x)\,, dx = \phi(a),$$

It extracts the test function's value at its center. This is the sampling, or sifting, property:

$$\int_{-\infty}^{\infty} \delta(x-a) f(x)\, dx = f(a).$$

For $x = a$, the distribution is supported at that point. For a smooth function, multiplication by the delta satisfies the distributional identity given below. This replaces the informal description of a function "becoming infinite" at $x = a$; integration against it selects the value at $x = a$. The product $f(x)\delta(x-a)$ is understood distributionally.

$$f(x)\delta(x-a) = f(a)\delta(x-a).$$

The delta is even:

$$\delta(-x) = \delta(x).$$

This follows by changing variables in its action on a test function. Another important property is scaling:

$$\delta(ax) = \frac{1}{|a|}\delta(x), \quad a \neq 0.$$

Consider

$$\int_{-\infty}^{\infty} \delta(ax) f(x) dx.$$

Use the substitution

$$u = ax \Rightarrow dx = \frac{du}{a}$$

For a negative scaling factor, the integration limits reverse; the Jacobian therefore contributes the reciprocal absolute value:

$$\int_{-\infty}^{\infty} \delta(ax) f(x) dx = \frac{1}{|a|}\int_{-\infty}^{\infty} \delta(u) f(u/a) du = \frac{f(0)}{|a|}.$$

The derivative of the delta, $\delta'(x)$, is defined by

$$\int_{-\infty}^{\infty} \delta'(x-a) f(x) dx = -f'(a)$$

This is useful in field theory and differential equations. Since $\delta(x)$ is a distribution, it is defined by its action on test functions $f(x)$; its derivative is not a pointwise derivative. The distributional derivative is defined through integration by parts, including the minus sign. To connect this definition with ordinary calculus, consider

$$\int_{-\infty}^{\infty} f(x)\, \delta'(x) dx$$

For a smooth regularization, ordinary integration by parts gives

$$\int u\, dv = uv - \int v\, du$$

$$\int_{-\infty}^{\infty} f(x)\delta_\varepsilon{}'(x) dx = [f(x)\delta_\varepsilon(x)]_{-\infty}^{\infty} - \int_{-\infty}^{\infty} f'(x)\delta_\varepsilon(x)\, dx.$$

$$[f(x)\delta_\varepsilon(x)]_{-\infty}^{\infty} = 0 \text{ for compactly supported test functions.}$$

$$\int_{-\infty}^{\infty} f(x)\delta'(x)\, dx = -\int_{-\infty}^{\infty} f'(x)\delta(x) dx = -f'(0).$$

For a positive width parameter, approximate the delta by symmetric functions $\delta_\epsilon(x)$ converging to $\delta(x)$ in the sense of distributions. A normalized Gaussian is

$$\delta_\epsilon(x) = \frac{1}{\epsilon\sqrt{\pi}} e^{-x^2/\epsilon^2}$$

Its derivative is

$$\delta_\epsilon'(x) = -\frac{2x}{\epsilon^{3\sqrt{\pi}}} e^{-x^2/\epsilon^2}$$

This regularized derivative is odd, has zero integral, and becomes concentrated near $x = 0$. Integrate against $f(x)$, then take $\epsilon \to 0$; integration by parts gives the rigorous distributional limit

$$\lim_{\epsilon \to 0} \int f(x)\, \delta_\epsilon'(x), dx = -f'(0).$$

For the translated delta $\delta(x-a)$,

$$\boxed{\int f(x)\,, \delta'(x-a), dx = -f'(a)}$$

The same argument applies after the substitution $y = x - a$. The following sequences represent the delta in the distributional limit, rather than as a pointwise limit:

a) Gaussian

$$\delta(x) = \lim_{\epsilon \to 0} \frac{1}{\sqrt{\pi\epsilon}} e^{-x^2/\epsilon}$$

b) Rectangular pulse

$$\delta_\varepsilon(x) = \frac{1}{2\varepsilon} \ (|x| < \varepsilon),$$
$$\delta_\varepsilon(x) = 0 \ (|x| \geq \varepsilon),$$
$$\varepsilon > 0, \qquad \delta_\varepsilon \to \delta \ (\varepsilon \to 0).$$

c) Cardinal sine

$$\delta(x) = \lim_{L \to \infty} \frac{\sin(Lx)}{\pi x}$$

These representations are used in analysis and numerical methods. The cardinal-sine representation is oscillatory and is not a nonnegative probability density; its convergence is understood against suitable test functions.

Example 1. Integral with a translated delta

$$\int_{-\infty}^{\infty} \delta(x-2) \cdot \cos(x)\, dx = \cos(2)$$

Example 2. Delta under a change of variables

$$\int_{-\infty}^{\infty} \delta(3x-6) \cdot f(x), dx = \frac{1}{3} f(2)$$

Apply the scaling property: $\delta(3x-6) = \frac{1}{3}\delta(x-2)$

Example 3. Derivative of the delta

$$\int_{-\infty}^{\infty} \delta'(x-1) \cdot (x^2+1) dx = -\frac{d}{dx}(x^2+1)|_{x=1} = -2$$

For localized particles, such as fixed ions, the nuclear charge density can be written as

$$\rho_n(\vec{r}) = \sum_\alpha Z_\alpha \delta\big(\vec{r} - \vec{R}_\alpha\big),$$

Here, Z_α is the nuclear charge number of the nucleus α at $\vec{R}_\alpha$. From this point onward, electronic-energy formulas use Hartree atomic units. The external potential $V_{ext}(\vec{r})$ acting on the electrons is

$$V_{ext}(\vec{r}) = \sum_{\alpha} \frac{-Z_{\alpha}}{|\vec{r} - \vec{R}_{\alpha}|}$$

Integration against the electron number density gives the electron–nucleus interaction energy:

$$\int \rho(\vec{r}) V_{ext}(\vec{r}) d\vec{r}.$$

11 Thomas–Fermi Kinetic-Energy Functional

In 1927, Thomas and Fermi developed a kinetic-energy approximation based locally on a uniform, noninteracting electron gas. For an unpolarized gas in three dimensions, in Hartree atomic units,

$$T_{TF} = C_F \int \rho^{5/3}(\boldsymbol{r}) d\boldsymbol{r},$$

where $C_F \approx 2.871$, more precisely as given below:

$$C_F = \frac{3}{10}(3\pi^2)^{2/3} \approx 2.871234.$$

The total electronic-energy functional is

$$E_{TF}[\rho(\boldsymbol{r})] = C_F \int \rho^{5/3}(\boldsymbol{r}) d\boldsymbol{r} - \int \rho(\boldsymbol{r}) v(\boldsymbol{r}) d\boldsymbol{r} + \frac{1}{2} \iint \frac{\rho(\boldsymbol{r})\rho(\boldsymbol{r}')}{|\boldsymbol{r} - \boldsymbol{r}'|} d\boldsymbol{r} d\boldsymbol{r}'. \quad (21)$$

The terms are the Thomas–Fermi kinetic energy, electron–nucleus attraction, and classical electron–electron Coulomb repulsion. Here, $v(\boldsymbol{r})$ denotes the positive magnitude of the attractive nuclear potential, so its contribution enters Equation **Error! Reference source not found.** with a minus sign. For one nucleus,

$$v(\boldsymbol{r}) = \frac{Z}{|\boldsymbol{r} - \boldsymbol{R}|}.$$

where $\boldsymbol{R}$ is the nuclear position. This was an early orbital-free approach to electronic structure. It captures some bulk trends but poorly describes atomic and molecular structure; the approximate kinetic energy is a major limitation. The original model also omits exchange and correlation, so we should not attribute its errors to kinetic energy alone. For now, the aim is to illustrate the functional derivatives of the terms in Equation **Error! Reference source not found.**.

The derivative of the Thomas–Fermi kinetic-energy term in Equation **Error! Reference source not found.**,

$$T_{TF}[\rho] = C_F \int \rho^{5/3}(\boldsymbol{r}) d\boldsymbol{r} = \int C_F \rho^{5/3}(\boldsymbol{r}) d\boldsymbol{r}$$

follows directly from Equation (18). With $t_{TF} = C_F \rho^{5/3}(\boldsymbol{r})$,

$$\frac{\delta T_{TF}[\rho]}{\delta\rho} = \frac{\partial t_{TF}}{\partial\rho} - \nabla\cdot\left(\frac{\partial t_{TF}}{\partial\nabla\rho}\right) = \frac{5C_F\rho^{2/3}(\boldsymbol{r})}{3}.$$

The gradient term vanishes because the kinetic-energy integrand depends on the density but not its gradient.

11.1 Electron–Nucleus Potential-Energy Functional

Define the positive attraction-magnitude functional appearing with a minus sign in Equation **Error! Reference source not found.**:

$$V[\rho(\boldsymbol{r})] = \int \rho(\boldsymbol{r})v(\boldsymbol{r})d\boldsymbol{r} = Z\int \frac{\rho(\boldsymbol{r})}{|\boldsymbol{r}-\boldsymbol{R}|}d\boldsymbol{r}, \tag{22}$$

Here, $|\boldsymbol{r}-\boldsymbol{R}|$ is the distance to the nucleus of atomic number Z, with

$$v(\boldsymbol{r}) = \frac{Z}{|\boldsymbol{r}-\boldsymbol{R}|}.$$

The derivative of Equation **Error! Reference source not found.** follows from Equation (18) or directly from the definition. To illustrate the latter,

$$\left.\frac{dV}{d\epsilon}\right|_{\epsilon=0} = \lim_{\epsilon\to0}\frac{V[\rho(r)+\epsilon\phi(r)]-V[\rho(r)]}{\epsilon}$$

$$= \lim_{\epsilon\to0}\frac{1}{\epsilon}\left\{\int[\rho(\boldsymbol{r})+\epsilon\phi(\boldsymbol{r})]v(\boldsymbol{r})\,d\boldsymbol{r} - \int\rho(\boldsymbol{r})v(\boldsymbol{r})\,d\boldsymbol{r}\right\}$$

$$= \lim_{\epsilon\to0}\frac{1}{\epsilon}\left\{\cancel{\int\rho(\boldsymbol{r})v(\boldsymbol{r})\,d\boldsymbol{r}} + \int\epsilon\phi(\boldsymbol{r})v(\boldsymbol{r})\,d\boldsymbol{r} - \cancel{\int\rho(\boldsymbol{r})v(\boldsymbol{r})\,d\boldsymbol{r}}\right\}$$

$$= \lim_{\epsilon\to0}\frac{1}{\epsilon}\int\epsilon v(\boldsymbol{r})\phi(\boldsymbol{r})\,d\boldsymbol{r} = \int v(\boldsymbol{r})\phi(\boldsymbol{r})\,d\boldsymbol{r}$$

$$\left.\frac{dV}{d\epsilon}\right|_{\epsilon=0} = \int v(\boldsymbol{r})\phi(\boldsymbol{r})\,d\boldsymbol{r} \tag{23}$$

Equation **Error! Reference source not found.** has the form

$$\left.\frac{dV}{d\epsilon}\right|_{\epsilon=0} = \int\frac{\delta V}{\delta\rho(\boldsymbol{r})}\phi(\boldsymbol{r})d\boldsymbol{r}$$

Therefore, the derivative of $V[\rho(\boldsymbol{r})]$ with respect to $\rho(\boldsymbol{r})$ is

$$\frac{\delta V}{\delta\rho(\boldsymbol{r})} = v(\boldsymbol{r}).$$

Because this functional is the positive attraction magnitude, the physical electron–nucleus energy is its negative, and its density derivative is the negative nuclear-potential magnitude.

11.2 Electron–Electron Coulomb-Energy Functional

The classical electron–electron interaction is described by the Coulomb, or Hartree, functional:

$$J[\rho] = \frac{1}{2}\iint \frac{\rho(\boldsymbol{r})\rho(\boldsymbol{r}')}{|\boldsymbol{r}-\boldsymbol{r}'|}\,d\boldsymbol{r}d\boldsymbol{r}'. \tag{24}$$

To differentiate this quadratic functional, first perturb both density factors. The first-order terms are

$$\delta J = \frac{1}{2}\int\int \frac{\delta\rho(\boldsymbol{r})\rho(\boldsymbol{r}') + \rho(\boldsymbol{r})\delta\rho(\boldsymbol{r}')}{|r-r'|}\,d\boldsymbol{r}\,d\boldsymbol{r}'.$$

By exchanging the two integration variables and using symmetry of the Coulomb kernel, the two contributions are equal. Thus,

$$\delta J = \int \delta\rho(\boldsymbol{r})\left[\int \frac{\rho(\boldsymbol{r}')}{|\boldsymbol{r}-\boldsymbol{r}'|}\,d\boldsymbol{r}'\right]d\boldsymbol{r}.$$

Equivalently, define the Hartree potential as

$$v_H(\boldsymbol{r}) = \int \frac{\rho(\boldsymbol{r}')}{|\boldsymbol{r}-\boldsymbol{r}'|}\,d\boldsymbol{r}'.$$

The first variation is its pairing with the density variation:

$$\delta J = \int v_H(\boldsymbol{r})\delta\rho(\boldsymbol{r})\,d\boldsymbol{r}\,.$$

Comparing this expression with the definition of the functional derivative,

$$\delta J = \int \frac{\delta J}{\delta\rho(\boldsymbol{r})}\delta\rho(\boldsymbol{r})\,d\boldsymbol{r},$$

gives

$$\frac{\delta J}{\delta\rho(\boldsymbol{r})} = \int \frac{\rho(\boldsymbol{r}')}{|\boldsymbol{r}-\boldsymbol{r}'|}\,d\boldsymbol{r}'. \tag{25}$$

The derivative in Equation (25), evaluated at a specified point, is itself a functional of the density. Differentiating once more gives the second functional derivative of Equation (24):

$$\frac{\delta^2 J}{\delta\rho(\boldsymbol{r}')\delta\rho(\boldsymbol{r})} = \int \frac{\delta(\boldsymbol{s}-\boldsymbol{r}')}{|\boldsymbol{r}-\boldsymbol{s}|}\,d\boldsymbol{s} = \frac{1}{|\boldsymbol{r}-\boldsymbol{r}'|}.$$

This follows from the delta identity kernel and integration over the primed variable. The Coulomb kernel is symmetric, so the second derivative is symmetric under interchange of its spatial arguments. More generally, equality of mixed functional derivatives requires appropriate differentiability; at coincident points, the singular kernel is interpreted in its integral or distributional sense.

11.3 von Weizsäcker Kinetic-Energy Functional

In 1935, von Weizsäcker introduced a density-gradient expression for electronic kinetic energy, motivating corrections to the local Thomas–Fermi approximation. Its full coefficient is exact for a single spatial orbital with constant phase; it is not the same as the second-order gradient-expansion correction for a slowly varying many-electron density. The functional is

$$T_w[\rho] = \frac{1}{8}\int \frac{\nabla\rho(\boldsymbol{r})\cdot\nabla\rho(\boldsymbol{r})}{\rho(\boldsymbol{r})}\,d\boldsymbol{r} = \int t_w d\boldsymbol{r},$$

where

$$t_w = \frac{1}{8}\frac{\nabla\rho\cdot\nabla\rho}{\rho} \qquad \text{and} \quad \rho = \rho(\boldsymbol{r}).$$

Here, we assume the density is positive wherever the derivative is evaluated, and we assume the boundary terms vanish. Behavior at density nodes requires a suitable limiting or weak interpretation. Using Equation (18), for a positive, sufficiently smooth density,

$$\frac{\partial\, t_w}{\partial\rho} = -\frac{1}{8}\frac{|\nabla\rho|^2}{\rho^2}, \qquad \frac{\partial\, t_w}{\partial(\nabla\rho)} = \frac{1}{4}\frac{\nabla\rho}{\rho},$$
$$\nabla\cdot\left(\frac{\nabla\rho}{\rho}\right) = \frac{\nabla^2\rho}{\rho} - \frac{|\nabla\rho|^2}{\rho^2},$$
$$\frac{\delta\, T_w}{\delta\rho} = -\frac{|\nabla\rho|^2}{8\rho^2} - \frac{1}{4}\left(\frac{\nabla^2\rho}{\rho} - \frac{|\nabla\rho|^2}{\rho^2}\right).$$

which gives

$$\frac{\delta T_w}{\delta\rho(\boldsymbol{r})} = \frac{1}{8}\frac{\nabla\rho\cdot\nabla\rho}{\rho^2} - \frac{1}{4}\frac{\nabla^2\rho}{\rho}.$$

12 Kinetic-Energy Density and Total Kinetic Energy

The following derivation relates the Laplacian form, involving ∇^2, to the gradient form, involving ∇, which defines the nonnegative kinetic-energy density $\tau(r)$. For the noninteracting system, let $T_s[\{\psi_i\}]$ denote the total kinetic energy. The sum below runs over occupied spin orbitals with unit occupation, with spin summation understood; for spatial orbitals, include their occupation numbers explicitly. In Hartree atomic units,

$$\tau(\mathbf{r}) = \frac{1}{2}\sum_i |\,\nabla\psi_i(\mathbf{r})|^2 \quad \text{and} \quad T_s[\{\psi_i\}] = -\frac{1}{2}\sum_i \int \psi_i^*(\mathbf{r})\nabla^2\psi_i(\mathbf{r})\,d\mathbf{r}$$

Use the divergence product rule $\nabla\cdot(f\mathbf{A}) = (\nabla f)\cdot\mathbf{A} + f(\nabla\cdot\mathbf{A})$, with f identified as the conjugate orbital ψ_i^* and $\mathbf{A}$ as its gradient $\nabla\psi_i$:

$$\nabla\cdot(\psi_i^*\nabla\psi_i) = \nabla\psi_i^*\cdot\nabla\psi_i + \psi_i^*\nabla^2\psi_i$$

Isolating the Laplacian term gives

$$\psi_i^*\nabla^2\psi_i = \nabla\cdot(\psi_i^*\nabla\psi_i) - \nabla\psi_i^*\cdot\nabla\psi_i.$$

Substitution into the total kinetic energy gives

$$T_s[\{\psi_i\}] = -\frac{1}{2}\sum_i \left(\int \nabla \cdot (\psi_i^* \nabla \psi_i)\, d\mathbf{r} - \int \nabla \psi_i^* \cdot \nabla \psi_i\, d\mathbf{r} \right).$$

Apply Gauss's divergence theorem to the first integral. It converts the volume integral to flux through the boundary surface S:

$$\int_V (\nabla \cdot \mathbf{F})\, dV = \oint_S \mathbf{F} \cdot d\mathbf{A}.$$

For localized bound states, the orbitals and their derivatives decay sufficiently rapidly as $r \to \infty$, so the surface flux of $\psi_i^* \nabla \psi_i$ vanishes. The same cancellation holds under suitable homogeneous or periodic boundary conditions; for periodic systems, opposite-face fluxes cancel. Only the gradient contribution remains, giving

$$T_s[\{\psi_i\}] = \frac{1}{2}\sum_i \int |\nabla \psi_i|^2\, d\mathbf{r} = \int \tau(\boldsymbol{r})\, d\mathbf{r}.$$

Here,

$T_s[\{\psi_i\}]$ is the noninteracting Kohn–Sham kinetic energy.
ψ_i are occupied Kohn–Sham spin orbitals.
$\nabla \psi_i$ is the orbital gradient.
$\tau(\boldsymbol{r})$ is the kinetic-energy density.
The local function $\tau(\boldsymbol{r})$ describes the spatial distribution of noninteracting kinetic energy in the Kohn–Sham system:

$$\tau(r) = \frac{1}{2}\sum_i^N |\nabla \psi_i(\mathbf{r})|^2$$

Conventions sometimes define twice $\tau(r)$, moving the explicit factor $1/2$ into the relation to total energy; this is a convention, not a consequence of atomic units alone. The quantity $\tau(r)$ is an ingredient of meta-GGA exchange–correlation functionals, including TPSS, SCAN, and members of the M06 family. Together with the density and its derivatives, it helps distinguish electronic environments. It also enters the electron localization function (ELF), which compares a Pauli excess kinetic-energy density with a uniform-gas reference, rather than using the raw ratio of $\tau(r)$ to the uniform-gas kinetic-energy density alone.

A useful property of $\tau(r)$ is nonnegativity: the squared moduli $|\ldots|^2$ ensure $\tau(r)$ is ≥ 0, including the possibility of zero. The alternative Laplacian form $\psi^* \nabla^2 \psi$ is not pointwise nonnegative. Under appropriate boundary conditions, the two forms integrate to the same energy, but they differ locally by a divergence. Thus, $\tau(\boldsymbol{r})$ provides a convenient nonnegative local descriptor; universal smoothness or numerical superiority does not follow from positivity alone.

12.1 Derivative of non-interacting Kohn-Sham energy functional

Because the orbitals may be complex, we treat ψ_i and ψ_i^* as independent variables when taking functional derivatives of

$$T_s[\{\psi_i\}] = -\frac{1}{2}\sum_i \int_\Omega \psi_i^*(\boldsymbol{r})\,\nabla^2\psi_i(\boldsymbol{r})\,d\boldsymbol{r}.$$

First, make the substitutions

$$\psi_i \to \psi_i + \epsilon\eta_i, \qquad \psi_i^* \to \psi_i^* + \epsilon\eta_i^*,$$

where η_i is an arbitrary admissible variation. Expanding to first order in ϵ,

$$T_s[\psi + \epsilon\eta] = T_s[\psi] + \epsilon\,\delta T_s + O(\epsilon^2),$$

with

$$\delta T_s = -\frac{1}{2}\sum_i \int_\Omega [\eta_i^*\nabla^2\psi_i + \psi_i^*\nabla^2\eta_i]\,d\boldsymbol{r}.$$

To identify a functional derivative, the variation must appear without spatial derivatives. Green's identity gives

$$\int_\Omega \psi_i^*\,\nabla^2\eta_i\,d\boldsymbol{r} = \int_\Omega \eta_i\,\nabla^2\psi_i^*\,d\boldsymbol{r} + \oint_{\partial\Omega}(\psi_i^*\,\partial_n\eta_i - \eta_i\,\partial_n\psi_i^*)\,dS.$$

Assume boundary conditions that make the surface term vanish, for example, sufficient decay at infinity, periodic boundary conditions, or homogeneous Dirichlet conditions on both orbitals and admissible variations. Then

$$\delta T_s = \sum_i \int_\Omega \left[\left(-\frac{1}{2}\nabla^2\psi_i\right)\eta_i^* + \left(-\frac{1}{2}\nabla^2\psi_i^*\right)\eta_i\right]d\boldsymbol{r}.$$

By definition,

$$\delta T_s = \sum_i \int_\Omega \left[\frac{\delta T_s}{\delta\psi_i^*(\boldsymbol{r})}\,\eta_i^*(\boldsymbol{r}) + \frac{\delta T_s}{\delta\psi_i(\boldsymbol{r})}\,\eta_i(\boldsymbol{r})\right]d\boldsymbol{r}.$$

Comparing coefficients,

$$\frac{\delta T_s}{\delta\psi_i^*(\boldsymbol{r})} = -\frac{1}{2}\nabla^2\psi_i(\boldsymbol{r}) \qquad\qquad \frac{\delta T_s}{\delta\psi_i(\boldsymbol{r})} = -\frac{1}{2}\nabla^2\psi_i^*(\boldsymbol{r}).$$

Thus, the derivative with respect to the conjugate orbital is the kinetic-energy operator

$$\hat{t} = -\frac{1}{2}\nabla^2$$

acting on that orbital. If the orbitals are restricted to be real, $\psi_i^* = \psi_i$, there is only one independent variation. The two contributions combine, giving

$$\frac{\delta T_s}{\delta \psi_i(\boldsymbol{r})} = -\nabla^2 \psi_i(\boldsymbol{r}) \quad \text{for real orbitals.}$$

The factor two comes from varying both occurrences of the same real field.

Exercises

1. Exchange–correlation energy remains a central challenge in Kohn–Sham density functional theory. The local density approximation (LDA) takes the form
$$E_{xc}[\rho(\boldsymbol{r})] = \int \varepsilon_{xc}\big(\rho(\boldsymbol{r})\big)\rho(\boldsymbol{r})d\boldsymbol{r},$$
where $\varepsilon_{xc}\big(\rho(\boldsymbol{r})\big)$ is the exchange–correlation energy per electron of a uniform electron gas at density $\rho(\boldsymbol{r})$. Find the functional derivative.
2. In Hartree–Fock theory, the classical Coulomb contribution is
$$E_C[\chi] = \frac{1}{2}\sum_{a=1}^{n}\sum_{b=1}^{n}\left\langle \chi_a(1)\chi_b(2)\left|\frac{1}{r_{12}}\right|\chi_a(1)\chi_b(2)\right\rangle,$$
Here, n is the number of electrons, χ are spin orbitals, and r_{12} is the interelectronic distance; the particle labels include spatial and spin coordinates. Find the derivative with respect to the conjugate spin orbital χ_k^*, treating orbitals and their conjugates as independent variables.
3. In Thomas–Fermi theory, the total electronic energy of an atom is the density functional
$$E[\rho] = C_F\int \rho^{5/3}(\boldsymbol{r})d\boldsymbol{r} - Z\int\frac{\rho(\boldsymbol{r})}{\boldsymbol{r}}d\boldsymbol{r} + \frac{1}{2}\int\int\frac{\rho(\boldsymbol{r})\rho(\boldsymbol{r}')}{|\boldsymbol{r}-\boldsymbol{r}'|}d\boldsymbol{r}\,d\boldsymbol{r}'.$$
Make this functional stationary subject to the particle-number constraint
$$\int \rho(\boldsymbol{r})d\boldsymbol{r} = n,$$
where n is the total number of electrons, Z is the atomic number, and C_F is the Thomas–Fermi constant. Require a nonnegative density and examine the conditions for a minimum.
4. Find the derivative of the following functional, where the vector square is the squared Euclidean norm:
$$E[\rho] = \int \frac{|\nabla\rho(r)|^2}{\rho(r)^{8/3}}\,dr.$$